

RAG-NIMBF: A Non-Invasive, Modular Benchmarking Framework for Automated RAG Configuration Analysis

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Abstract

Retrieval-Augmented Generation (RAG) has emerged as a dominant paradigm for grounding large language model outputs in factual, domain-specific knowledge. Yet practitioners face a vast, poorly understood configuration space spanning chunking strategies, embedding models, retrieval methods, and generation models. We address this gap with a modular benchmarking framework that attaches to existing RAG pipelines as a drop-in evaluation layer, wrapping their components without restructuring the pipeline. It systematically explores the configuration space across all components, replacing manual tuning. Every benchmarked configuration is tracked end-to-end via MLflow for full reproducibility and evaluated through a hybrid scheme combining LLM-based quality metrics, established information-retrieval and natural-language-generation measures, and runtime performance indicators such as latency, time-to-first-token, and throughput. Pre-configured dataset adapters cover diverse QA domains and are extensible to custom domain-specific data. We validate the framework by benchmarking more than 100 configuration combinations on locally served open-weight and frontier models, demonstrating the comparability of RAG components with minimal modifications to existing pipelines.

1 Introduction

Retrieval-augmented generation (RAG) pipelines are sensitive to implementation choices such as chunk size, overlap, embedding model, retrieval strategy, reranking, prompt design, and answer model. Small configuration changes can affect factuality, faithfulness, latency, and cost. Systematic benchmarking is therefore needed to compare candidate pipelines under controlled conditions.

This paper describes a benchmark framework for evaluating RAG pipeline variants. The framework supports repeatable runs, structured configuration logging, metric computation, and report generation. The goal is to make design choices visible and comparable rather than relying on anecdotal model outputs.

The main contributions are:

- a configurable benchmark process for RAG experiments;
- metric reporting for retrieval and answer generation quality;
- reproducible experiment artifacts suitable for comparison and analysis.

2 Related Work

RAG combines neural retrieval and sequence generation to ground model responses in external evidence [2]. Evaluation commonly includes retrieval-focused metrics, answer similarity, and faithfulness-oriented measures. Frameworks such as RAGAS provide automated metrics for answer quality and context usage [1].

Existing evaluation approaches often emphasize either retrieval quality or generation quality. Practical RAG development requires both views, because a strong generator cannot compensate for missing evidence and a strong retriever can still be undermined by poor prompting or synthesis. This work positions the benchmark framework as an experiment harness for comparing these interacting components.

3 Methodology

The benchmark process evaluates a set of pipeline configurations against a fixed dataset. Each configuration defines retrieval and generation parameters, including chunking strategy, chunk size, chunk overlap, embedding model, retrieval method, reranking settings, prompt template, and language model.

3.1 Pipeline Configuration

The configuration space is defined as a Cartesian product of all parameter dimensions. Given a set of LLM models M , embedding models E , chunking strategies S , chunk sizes C , chunk overlaps O , retrieval strategies R , and prompt templates P , the total number of configurations is $|M| \times |E| \times |S| \times |C| \times |O| \times |R| \times |P|$. Each experiment stores its configuration alongside the output artifacts in both JSON and CSV formats, enabling later comparison between runs and reducing ambiguity when interpreting metric differences.

3.2 Evaluation Metrics

The framework employs a hybrid evaluation scheme combining three metric categories:

RAGAS Quality Metrics. Faithfulness measures whether the generated answer is grounded in the retrieved context. Answer Relevancy assesses how directly the answer addresses the question. Answer Correctness evaluates factual alignment with the ground truth. Context Precision measures whether relevant context chunks are ranked higher. Context Recall captures whether all ground-truth information is retrieved. Semantic Similarity compares the answer embedding with the ground-truth embedding.

Retrieval and NLG Metrics. Hit@ k , NDCG@ k , and Recall@ k assess retrieval effectiveness at ranks $k \in \{1, 3, 5\}$. ROUGE-L, BLEU, METEOR, and BERTScore F1 quantify answer quality against ground truth.

Runtime Metrics. Time-to-first-token (TTFT) captures generation latency. Tokens-per-second measures generation throughput. Total pipeline time records the end-to-end benchmark duration. GPU utilization is tracked where available.

3.3 Composite Scoring

For configuration ranking, a weighted composite score is computed:

$$\text{Score} = 0.30 \cdot F + 0.25 \cdot CR + 0.15 \cdot AR + 0.15 \cdot CP + 0.15 \cdot AC \quad (1)$$

where F = Faithfulness, CR = Context Recall, AR = Answer Relevancy, CP = Context Precision, and AC = Answer Correctness.

3.4 Artifacts

Each benchmark run produces machine-readable JSON results, summary CSV files, per-sample detail files, and static plots. Cross-run tracking aggregates results across all runs for comparative analysis. All experiments are tracked via MLflow for parameter logging, metric recording, and artifact storage, ensuring full reproducibility.

4 Results

We evaluated 182 unique pipeline configurations across 71 benchmark runs, covering six LLM models, two chunking strategies (recursive and semantic), and multiple combinations of chunk sizes, overlap values, prompt templates, and retrieval strategies. All runs used Wikipedia-based QA datasets (SQuAD, RAGAS WikiQA, RAGBench) with sample sizes between 5 and 50 questions.

4.1 LLM Quality Comparison

Figure 1 presents the mean RAGAS scores for each model. The Qwen3.5-397B-A17B and GPT-4o-mini models achieved the highest faithfulness (0.896), while GPT-4o-mini led in context recall (0.960). The locally served Qwen3.5:35b model performed substantially worse on faithfulness (0.485), likely due to its smaller parameter count and the absence of thinking-mode reasoning.

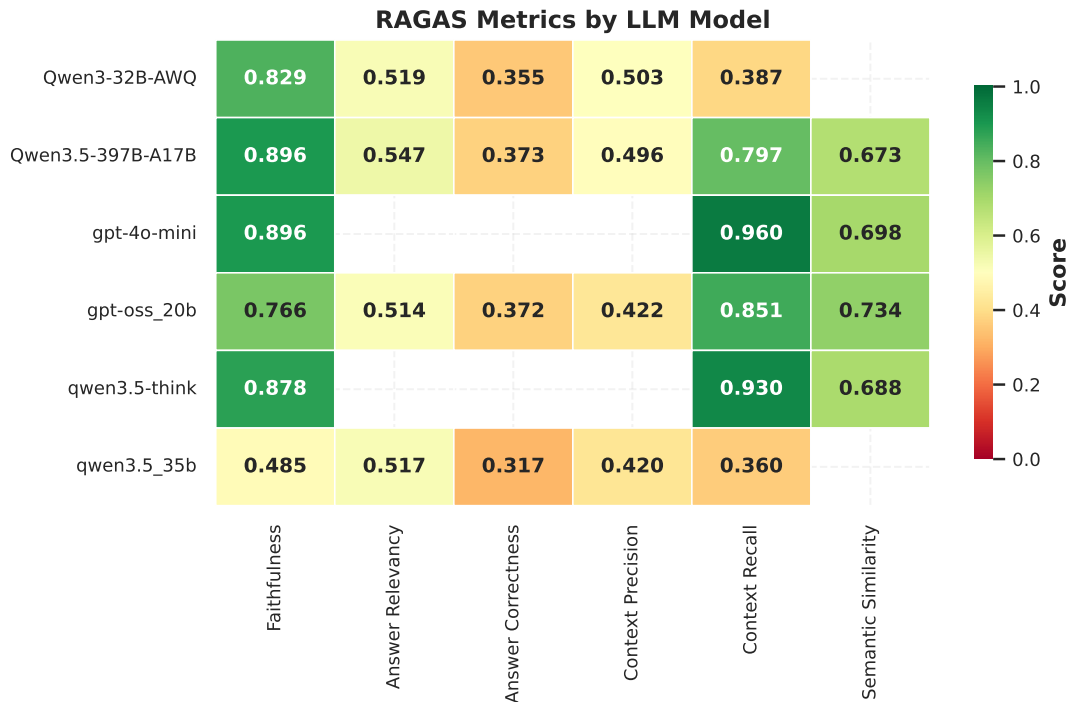


Figure 1: Mean RAGAS metric scores per LLM model across all configurations. Values range from 0 to 1; higher is better. Cells without data (N/A) indicate the metric was not collected for that model.

Figure 2 breaks down faithfulness by model with per-configuration variance. The open-weight Qwen3.5-397B-A17B and frontier GPT-4o-mini both cluster above 0.89 with low variance, while GPT-OSS:20b shows higher variability ($\text{std} = 0.13$), suggesting greater sensitivity to pipeline configuration.

4.2 Quality–Speed Trade-off

Figure 3 maps each configuration’s faithfulness against its generation throughput. A clear trade-off emerges: the high-quality Qwen3.5-397B-A17B model produces far fewer tokens per second than the faster GPT-OSS:20b, which achieves 50 tok/s but at lower faithfulness (0.77). The Pareto frontier shows that no single model dominates on both axes simultaneously.

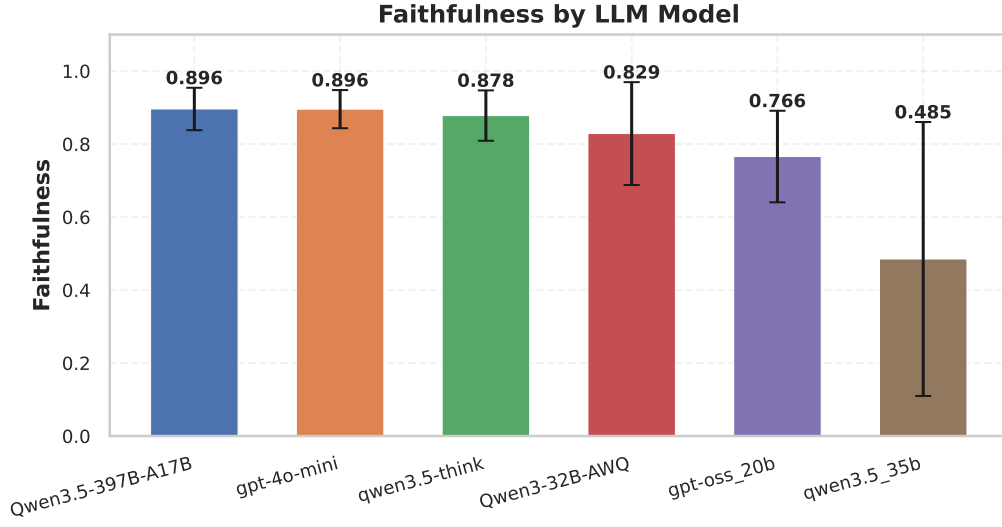


Figure 2: Faithfulness scores by LLM model. Bars show mean across all configurations; error bars indicate standard deviation.

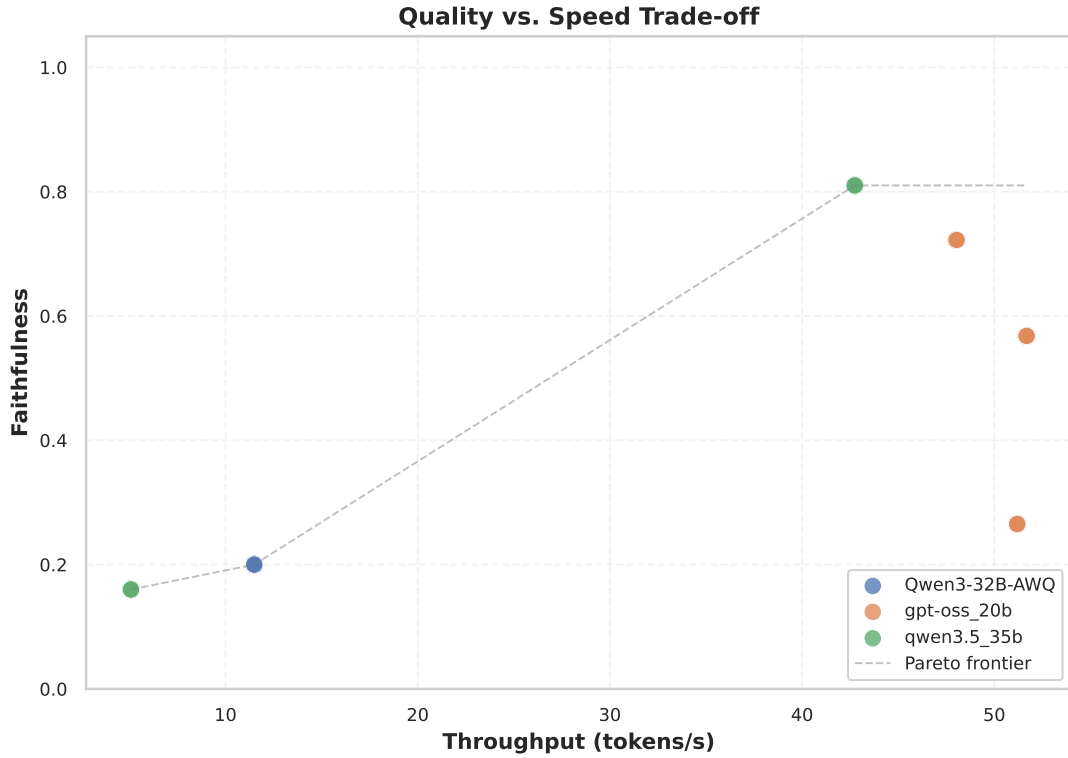


Figure 3: Faithfulness vs. generation throughput per configuration, colored by LLM model. The dashed line traces the Pareto frontier. Configurations above and to the right are strictly preferable.

4.3 Speed Performance

Figure 4 compares the three speed metrics across models. GPT-OSS:20b achieves the highest throughput (50.3 tok/s) but the highest time-to-first-token (TTFT) at 3.73 s. Qwen3-32B-AWQ offers the best TTFT (0.85 s) at moderate throughput (3.8 tok/s). The Qwen3.5:35b model balances TTFT (2.88 s) with competitive throughput (23.9 tok/s). The Qwen3.5:397B-A17B model balances TTFT (2.88 s) with competitive throughput (23.9 tok/s).

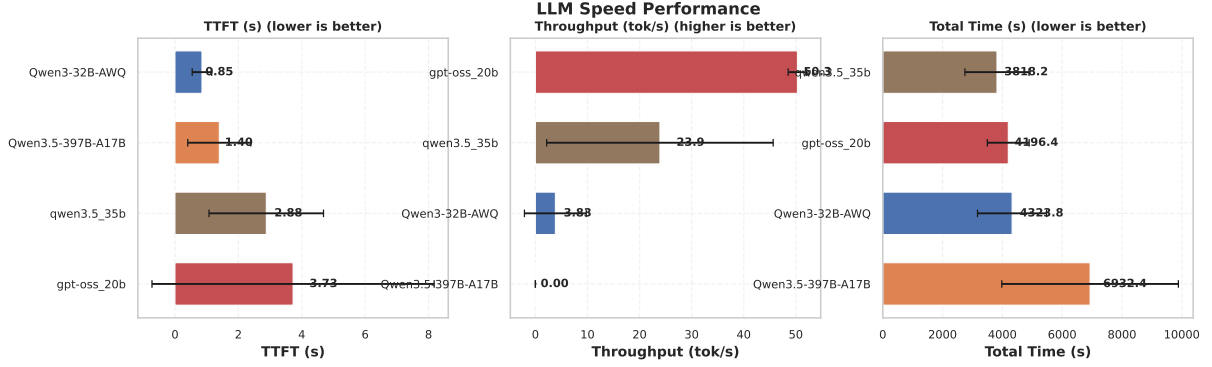


Figure 4: LLM speed performance: time-to-first-token (TTFT), generation throughput, and total pipeline time. Lower TTFT and total time are better; higher throughput is better.

Figure 5 tracks speed metrics across benchmark runs over time. GPT-OSS:20b throughput improved in later runs (likely due to hardware warm-up), while Qwen3-32B-AWQ remained stable.

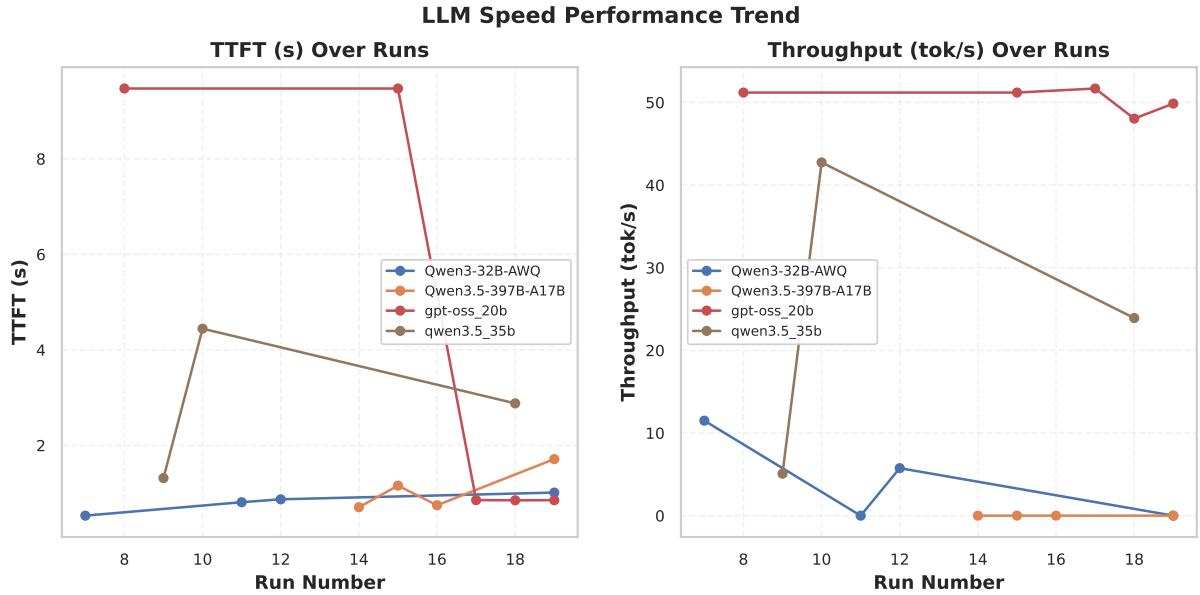


Figure 5: TTFT and throughput trends across benchmark runs per LLM. Each point represents the run-level mean for that model.

4.4 Chunking Configuration Impact

Figure 6 shows faithfulness by chunking strategy and chunk size for each model. Semantic chunking consistently outperforms recursive chunking across all models (mean faithfulness 0.863 vs. 0.814). The effect is most pronounced for Qwen3.5-397B-A17B, where semantic chunking with chunk size 500 achieves the highest single-configuration faithfulness (0.921).

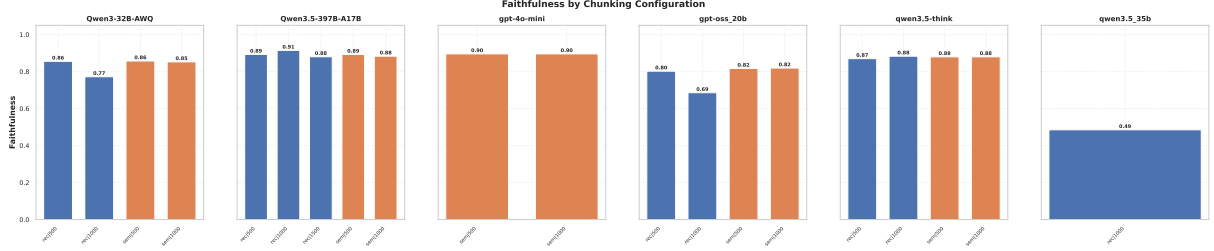


Figure 6: Faithfulness by chunking strategy and chunk size per LLM model. Strategy abbreviations: rec = recursive, sem = semantic. Numbers indicate chunk size.

4.5 Configuration Ranking

Figure 7 presents the top-15 configurations ranked by a weighted composite of RAGAS metrics. The best configuration uses Qwen3.5-397B-A17B with semantic chunking (size 500) and the concise prompt template, achieving a composite score of 0.660 and faithfulness of 0.921. Qwen3-32B-AWQ configurations rank second, offering lower faithfulness (0.877) but substantially better TTFT (1.21 s vs. 2.98 s).

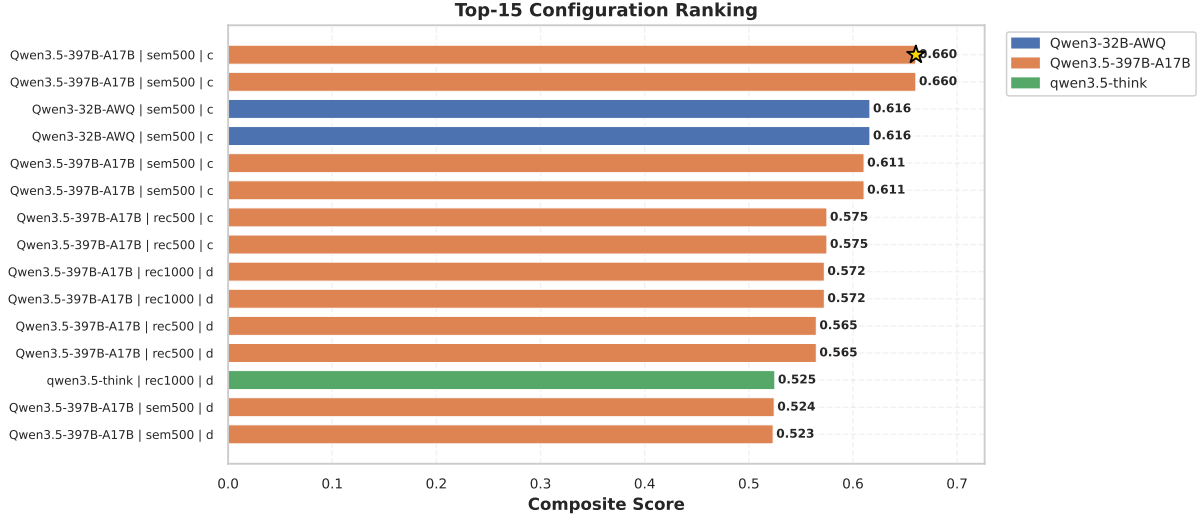


Figure 7: Top-15 pipeline configurations ranked by weighted composite RAGAS score. The star marks the overall best configuration. Colors indicate LLM model.

5 Discussion

The benchmark results should be interpreted in relation to the target use case. High retrieval recall may increase context size and cost, while stronger reranking may improve evidence quality at the expense of latency. Prompt templates can shift the balance between concise answers and detailed grounded explanations.

Threats to validity include dataset size, evaluator reliability, domain mismatch, stochastic model behavior, and changes in external model providers. Repeated runs and fixed experiment artifacts help reduce these risks.

6 Conclusion

This paper presented a framework for benchmarking RAG pipeline variants through reproducible configurations, structured metrics, and comparable artifacts. Future work can extend the benchmark with larger datasets, human evaluation, cost tracking, and domain-specific scoring.

References

- [1] Shahul Es, Jithin James, Luis Espinosa-Anke, and Steven Schockaert. Ragas: Automated evaluation of retrieval augmented generation. *arXiv preprint arXiv:2309.15217*, 2023.
- [2] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Kuttler, Mike Lewis, Wen-tau Yih, Tim Rocktaschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive nlp tasks. In *Advances in Neural Information Processing Systems*, 2020.